

Supporting Information

Rational Design of Hierarchical Zeolite Encapsulating FeMnK

Architecture to Enhance Light Olefins Selectivity in Fischer-Tropsch

Synthesis

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Tables and Figures

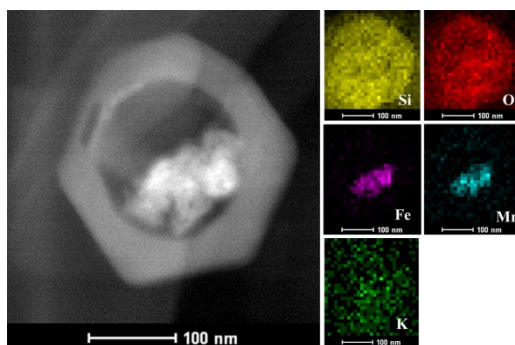


Figure S1. The HAADF-STEM images and the results of EDX elemental (Fe, Mn, K, Si, O) mapping of FeMnK@H-S-1(0.10) catalyst

Table S1. Element content of FeMnK@H-S-1(x) catalysts.

Catalyst	Fe (wt	Mn (wt	K (wt	Na (wt	Fe (wt
	%) ^a	%) ^a	%) ^a	%) ^a	%) ^b
FeMnK@H-S-1(0.00)	8.17	4.21	0.65	-	1.12
FeMnK@H-S-1(0.10)	7.81	4.08	0.56	0.26	1.03
FeMnK@H-S-1(0.20)	8.22	4.12	0.55	1.19	1.03
FeMnK@H-S-1(0.50)	9.19	4.59	0.53	1.33	1.06
FeMnK@H-S-1(0.67)	8.45	4.14	0.59	1.09	1.44
FeMnK@H-S-1(0.80)	9.20	4.66	0.48	1.78	1.20

^a Determined by XRF

^b Determined by XPS

Table S2. Textural properties of FeMnK@H-S-1(x) catalysts

Catalyst	S _{BET} ^a (m ² g ⁻¹)	S _{micro} ^b (m ² g ⁻¹)	S _{meso} ^c (m ² g ⁻¹)	V _{micro} ^b (cm ³ g ⁻¹)	V _{total} ^e (cm ³ g ⁻¹)	HF
FeMnK@H-S-1(0.00)	354.2	259.7	94.5	0.112	0.475	0.063
FeMnK@H-S-1(0.10)	321.1	218.5	102.6	0.096	0.471	0.065
FeMnK@H-S-1(0.20)	305.5	192.4	113.1	0.088	0.398	0.081
FeMnK@H-S-1(0.50)	300.4	209.6	90.8	0.092	0.376	0.074
FeMnK@H-S-1(0.67)	256.0	192.9	63.1	0.085	0.311	0.067
FeMnK@H-S-1(0.80)	216.2	156.8	59.4	0.071	0.283	0.069

^a BET method. ^b Measured by t-plot method.

^c Mesoporous surface area: S_{meso} = S_{BET} - S_{micro}.

^e Single point pore volume obtained from N₂ adsorbed at P/P₀ = 0.99.

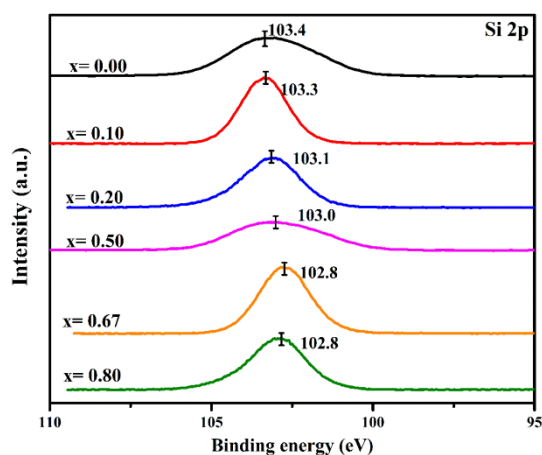


Figure S2. XPS spectra of Si 2p on different FeMnK@H-S-1(x) catalysts.

Table S3. Catalytic performance of FeMnK/S-1 catalyst

Catalyst	CO Conv. (%)	CO ₂ Sele. (%)	Hydrocarbon distribution (C				O/P
			%)				
			CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ^o -C ₄ ^o	C ₅ ⁺	
FeMnK/S-1	15.2	36.0	13.8	33.9	6.5	45.8	5.2

Reaction conditions: 280 °C, 0.5 MPa, 4000 h⁻¹, H₂/CO=1.90. m_(catalyst)=0.85 g

Table S4. Catalytic performance of FeMnK+H-S-1 and FeMnK catalysts

Catalyst	CO Conv. (%)	CO ₂ Sele. (%)	Hydrocarbon distribution (C				O/ P
			%)				
			CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ^o -C ₄ ^o	C ₅ ⁺	
FeMnK+H-S-1	13.3	47.8	16.9	37.1	8.3	37.6	4.5
FeMnK	11.8	47.2	16.8	32.0	7.4	43.8	4.3

Reaction conditions: 280 °C, 0.5 MPa, 4000 h⁻¹, H₂/CO=1.90.

Table S5. Comparison of the results in literature works for CO Hydrogenation to light olefins of iron-based catalysts and our work.

Catalyst	CO	FTY $\mu\text{mol}_{\text{CO}}\cdot\text{g}_{\text{Fe}}^{-1}\cdot\text{s}^{-1}$	CO ₂	Hydrocarbon distribution				STY
	Conv. (%)		Sele. (%)	CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ⁰ -C ₄ ⁰	C ₅ ⁺	$\mu\text{mol}_{\text{C}_2^+=\text{C}_4^+=}\cdot\text{g}_{\text{Fe}}^{-1}\cdot\text{s}^{-1}$
Fe/C@Si-SAPO·K ¹	N.A.	27.0	25.0	34.	40.3	13.6	11.	N.A.
FeMnK/Al ₂ O ₃ @S2 ²	4.0	28.0	25.1	42.	38.1	15.6	3.4	N.A.
Fe ₂₀ -Mn _{1.0} /SiO ₂ ³	14.4	N.A.	23.3	21.	32.3	N.A.	19.	N.A.
Fe/MnK-CNTs ⁴	22.7	N.A.	N.A.	28.	50.3	7.6	13.	N.A.
Fe-Na-S/CNF ⁵	N.A.	26.8	N.A.	36.	52.0	6.0	6.0	N.A.
Fe-K/NCNTs ⁶	16.5	27.9	23.6	17.	54.6	5.9	22.	N.A.
FeBi/CNT-in ⁷	25.6	47.0	37.3	27.	62.4	6.5	4.6	18.4
FePb/CNT-in ⁷	35.9	66.0	39.9	26.	58.9	7.4	7.6	23.4.
This work ⁸	13.9	85.9	38.6	24.	50.6	9.9	15.	26.7

1. Reaction conditions: 340 °C, 1.0 MPa, H₂/CO = 1, 16 SL.g_{cat}⁻¹.h⁻¹

2. Reaction conditions: 340 °C, 2.0 MPa, H₂/CO=1, WHSV=4500 mL.g⁻¹.h⁻¹

3. Reaction conditions: 260 °C 2.0 MPa, H₂/CO=1, GHSV=4000 h⁻¹

4. Reaction conditions: 270 °C 2.0 MPa, H₂/CO=1, GHSV=30000mL. g_{cat}⁻¹.h⁻¹

5. Reaction conditions: 350 °C, 0.1 MPa, H₂/CO = 1,

6. Reaction conditions: 300 °C, 0.1 MPa, $H_2/CO = 1$, $WHSV=4200 \text{ mL} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$
7. Reaction conditions: 350 °C, 0.1 MPa, $H_2/CO = 1$, $GHSV=3.4 \text{ L} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{h}^{-1}$
8. Reaction conditions: 280 °C, 0.5 MPa, $H_2/CO = 1.9$, $GHSV=4000 \text{ h}^{-1}$

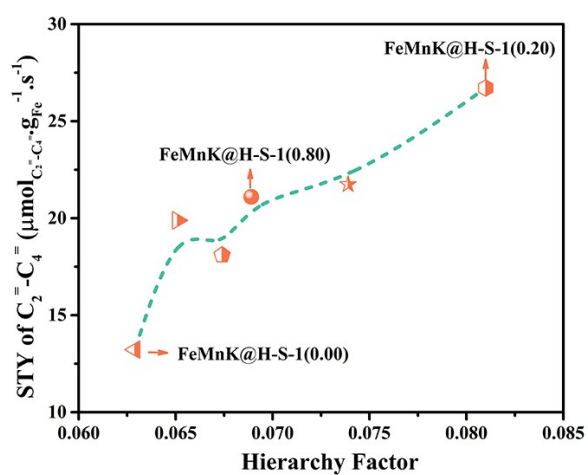


Figure S3. The space time yield (STY) of light olefins versus hierarchy factor.

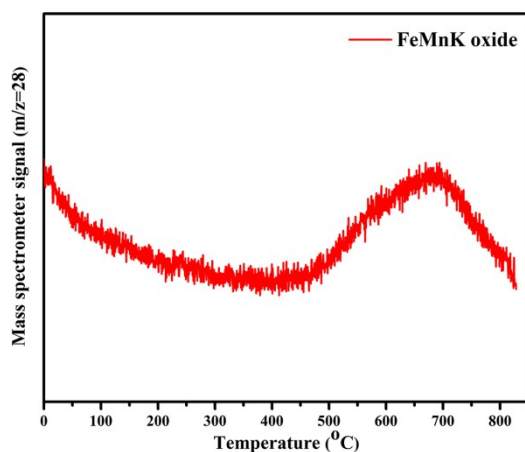


Figure S4. C₂H₄-TPD-MS profile of reduced FeMnK oxide.

Table S6. Element content of Na modified FeMnK@H-S-1(0.00) catalyst derived from XRF

Catalyst	Fe (wt %)	Mn (wt %)	K (wt %)	Na (wt %)
FeMnK@H-S-1(0.00)-1.17Na	7.49	3.60	0.69	1.17

Table S7. Catalytic performance of FeMnK@H-S-1(0.00) and Na modified FeMnK@H-S-1(0.00) catalysts

Catalyst	CO _{Conv.} (%)	CO ₂ Sele. (%)	Hydrocarbon distribution				O/ P
			(C%)				
			CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ⁰ -C ₄ ⁰	C ₅ ⁺	
FeMnK@H-S-1(0.00)	7.5	16.2	33.5	31.8	29.4	5.3	1.1
FeMnK@H-S-1(0.00) 1.17Na	7.8	17.3	31.8	33.9	27.8	6.5	1.2

Reaction conditions: 280 °C, 0.5 MPa, 4000 h⁻¹, H₂/CO=1.90. m_(catalyst)=0.85 g

